CBSE Class XII Board Examination 2027 — Expected Paper

CHEMISTRY (Theory) · Code 043

Time allowed: 3 hours **Maximum marks: 70**

General instructions

1. This paper has 33 questions in five sections, A–E. All questions are compulsory.
2. Section A: Q1–16 are MCQs of 1 mark each (Q13–16 are Assertion–Reason). Section B: Q17–21, 2 marks each. Section C: Q22–28, 3 marks each.
3. Section D: Q29–30 are case-based questions of 4 marks each. Section E: Q31–33, 5 marks each.
4. This is an original practice paper built on the latest CBSE sample-paper pattern. It is not an official CBSE paper.

Section A — Multiple choice (1 mark each). Questions 13–16 are Assertion–Reason.

1. Which concentration term does not depend on temperature? **[1]**

(a) Molarity (c) Molality
(b) Normality (d) Formality

2. The colligative property best suited to find the molar mass of proteins is **[1]**

(a) relative lowering of vapour pressure (c) depression in freezing point
(b) elevation in boiling point (d) osmotic pressure

3. The unit of the rate constant of a first-order reaction is **[1]**

(a) mol L^{-1} s^{-1} (c) L mol^{-1} s^{-1}
(b) s^{-1} (d) L^{2} mol^{-2} s^{-1}

4. On dilution, the molar conductivity of an electrolyte solution **[1]**

(a) increases (c) stays the same
(b) decreases (d) first decreases then increases

5. Which ion has the maximum number of unpaired electrons? **[1]**

(a) Ti^{3+} (c) Cr^{3+}
(b) V^{3+} (d) Mn^{2+}

6. The lanthanoid that shows a stable +4 oxidation state is **[1]**

(a) Ce (c) Gd
(b) Eu (d) Lu

7. The IUPAC name of $[Co(NH_3)_5Cl]Cl_2$ is **[1]**

(a) Pentaamminechloridocobalt(III) chloride (c) Pentaamminecobalt(III) chloride
(b) Chloridopentaamminecobalt(II) chloride (d) Pentaamminechloridocobalt(II) dichloride

8. Which is an ambidentate ligand? **[1]**

(a) NH_3
(b) en
(c) SCN^-
(d) H_2O

9. Which is most reactive towards SN2 substitution? **[1]**

(a) CH_3I
(b) CH_3CH_2I
(c) $(CH_3)_2CHI$
(d) $(CH_3)_3CI$

10. Phenol is more acidic than ethanol because **[1]**

(a) phenol has a higher molar mass
(b) the phenoxide ion is resonance-stabilised
(c) ethanol forms hydrogen bonds
(d) phenol is aromatic

11. The Lucas test is used to distinguish between **[1]**

(a) aldehydes and ketones
(b) primary, secondary and tertiary alcohols
(c) phenol and ethanol
(d) primary and secondary amines

12. Which compound gives a positive iodoform test? **[1]**

(a) Methanol
(b) Ethanol
(c) Propan-1-ol
(d) Benzyl alcohol

13. **Assertion (A):** Aniline does not undergo Friedel–Crafts reactions. **[1]**
Reason (R): Aniline forms a salt with the Lewis acid $AlCl_3$.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Transition metal ions are usually coloured. **[1]**
Reason (R): They have partially filled d-orbitals that allow d–d transitions.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Glucose gives a positive Tollens' test. **[1]**
Reason (R): The open-chain form of glucose contains an aldehyde (–CHO) group.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

16. **Assertion (A):** Alkyl halides have lower boiling points than the corresponding alkanes. **[1]**
Reason (R): The C–X bond in alkyl halides is polar.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Section B — Very short answer (2 marks each)

17. State Raoult's law for a solution of two volatile liquids. What is a positive deviation from it? Give one example. **[2]**

18. For $2A + B \rightarrow$ products, rate $= k[A][B]^2$. (i) What is the overall order? (ii) How does the rate change if [B] is doubled? **[2]**

19. Calculate ΔG° for the Daniell cell ($E^\circ_{cell} = 1.10$ V, $F = 96500$ C mol^{-1}). **[2]**

20. Explain: (a) Zn, Cd and Hg are not regarded as transition elements. (b) Cu^+ is unstable in aqueous solution. **[2]**

21. Write the chemical equation for (a) Rosenmund reduction and (b) Clemmensen reduction. **[2]**

Section C — Short answer (3 marks each)

22. 1.8 g of glucose is dissolved in 100 g of water. Calculate the freezing point of the solution ($K_f = 1.86$ K kg mol^{-1}). What is abnormal molar mass? **[3]**

23. Show that for a first-order reaction, the time for 99% completion is twice the time for 90% completion. **[3]**

24. Using valence bond theory, explain why $[Ni(CN)_4]^{2-}$ is square planar and diamagnetic while $[NiCl_4]^{2-}$ is tetrahedral and paramagnetic. **[3]**

25. Explain the SN1 mechanism for the hydrolysis of tert-butyl bromide. Why do optically active halides give racemic mixtures by SN1? **[3]**

26. How will you convert (a) phenol to salicylic acid, (b) propanone to propan-2-ol and (c) benzene to acetophenone? **[3]**

27. (a) Arrange in increasing order of basic strength in aqueous solution: $C_2H_5NH_2$, $(C_2H_5)_2NH$, $(C_2H_5)_3N$, NH_3. (b) Write the Hoffmann bromamide degradation reaction. (c) Why are aromatic diazonium salts more stable than aliphatic ones? **[3]**

28. (a) What is the primary structure of a protein? (b) What happens during denaturation? (c) Give two structural differences between DNA and RNA. **[3]**

Section D — Case-based questions (4 marks each)

29. **Case study: Kohlrausch's law.** At infinite dilution every ion migrates independently, and the limiting molar conductivity of an electrolyte equals the sum of the contributions of its ions. This lets us find Λ°_m for weak electrolytes. **[4]**

(i) Define limiting molar conductivity.

(ii) Calculate $\Lambda^\circ_m(CH_3COOH)$, given $\Lambda^\circ_m(HCl) = 426.2$, $\Lambda^\circ_m(NaCl) = 126.4$ and $\Lambda^\circ_m(CH_3COONa) = 91.0$ S cm^2 mol^{-1}.

(iii) Why does Λ_m of a weak electrolyte rise steeply on dilution?

(iv) Write the SI unit of conductivity.

30. **Case study: structure of glucose.** Several reactions of glucose establish its open-chain structure, while others it fails to give suggest a cyclic structure. **[4]**

(i) What does the formation of n-hexane with HI show?

(ii) Name the product with bromine water and what it shows.

(iii) What does the formation of a pentaacetate show?

(iv) Which observation supports a cyclic structure?

Section E — Long answer (5 marks each)

31. (a) Write the Nernst equation and calculate the emf of the cell Mg | Mg^{2+}(0.001 M) || Cu^{2+}(0.0001 M) | Cu, given E°cell = 2.71 V. **[5]**

(b) What is a fuel cell? Write the electrode reactions of the H_2–O_2 fuel cell.

32. (a) How is $KMnO_4$ prepared from pyrolusite? (b) Write the ionic equation for the oxidation of Fe^{2+} by acidified $K_2Cr_2O_7$. (c) What is lanthanoid contraction? Give two consequences. **[5]**

33. (a) Give chemical tests to distinguish (i) propanal from propanone and (ii) acetophenone from benzophenone. **[5]**

(b) Write the aldol condensation of ethanal.

(c) Arrange in increasing order of acidity: CH_3COOH, FCH_2COOH, $ClCH_2COOH$.

Marking scheme — Chemistry (043)

Value points and mark split

Section A — Multiple choice (1 mark each). Questions 13–16 are Assertion–Reason.

1. (c) Molality is based on mass, which does not change with temperature.

2. (d) Osmotic pressure: it is measured at room temperature and is large even for dilute solutions.

3. (b) s^{-1}.

4. (a) It increases: more ions per mole (weak electrolytes) and less interionic attraction.

5. (d) Mn^{2+} is $3d^5$, with 5 unpaired electrons.

6. (a) Ce^{4+} has the noble gas ($4f^0$) configuration.

7. (a) Pentaamminechloridocobalt(III) chloride.

8. (c) SCN^- can bind through S or N.

9. (a) CH_3I has the least steric hindrance.

10. (b) The negative charge of phenoxide is delocalised over the ring.

11. (b) Turbidity forms immediately for 3°, in about 5 min for 2°, and not at room temperature for 1° alcohols.

12. (b) Ethanol has the $CH_3CH(OH)–$ group.

13. (a) N becomes positively charged and deactivates the ring.

14. (a) Absorbed visible light promotes electrons between split d levels.

15. (a) The aldehyde group reduces Ag^+ to a silver mirror.

16. (d) Haloalkanes boil higher (greater mass and dipole forces), so A is false.

Section B — Very short answer (2 marks each)

17. p_A = x_A p°_A and p_B = x_B p°_B; p_total = p_A + p_B. Positive deviation: vapour pressure is higher than predicted because A–B interactions are weaker than A–A and B–B; ΔH_mix > 0 and ΔV_mix > 0. Example: ethanol + acetone. [1 + 1]

18. (i) 1 + 2 = 3. (ii) Rate ∝ $[B]^2$, so it becomes 4 times. [1 + 1]

19. $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, n = 2. $\Delta G° = -nFE° = -2 \times 96500 \times 1.10 = -212300$ J $mol^{-1} = -212.3$ kJ mol^{-1}. [2]

20. (a) They have completely filled d^{10} orbitals in the ground state and in their common oxidation states. (b) $Cu^{2+}(aq)$ has a much more negative hydration enthalpy, so $2Cu^+ \rightarrow Cu^{2+} + Cu$ (disproportionation) is favoured. [1 + 1]

21. (a) $RCOCl + H_2 \rightarrow RCHO + HCl$, over $Pd/BaSO_4$. (b) $R_2C{=}O + 4[H] \rightarrow R–CH_2–R + H_2O$, using Zn–Hg/conc. HCl. [1 + 1]

Section C — Short answer (3 marks each)

22. n = 1.8/180 = 0.01 mol; m = 0.01/0.1 = 0.1 mol kg^{-1}. $\Delta T_f = K_f m = 0.186$ K, so $T_f = 273.15 - 0.186 = 272.96$ K. Abnormal molar mass: a molar mass from colligative data that differs from the true value because the solute associates or dissociates (corrected with the van't Hoff factor i). [2 + 1]

23. $t = (2.303/k) \log([A]_0/[A])$. $t_{99} = (2.303/k) \log 100 = (2.303/k) \times 2$. $t_{90} = (2.303/k) \log 10 = (2.303/k) \times 1$. So $t_{99} = 2\, t_{90}$. [3]

24. Ni^{2+} is $3d^8$. CN^- is a strong-field ligand: it pairs the d electrons, leaving one 3d orbital empty, so the hybridisation is dsp^2 (square planar) with no unpaired electrons (diamagnetic). Cl^- is weak-field: no pairing occurs, so the hybridisation is sp^3 (tetrahedral) with 2 unpaired electrons (paramagnetic). [1½ + 1½]

25. Step 1 (slow): $(CH_3)_3C–Br \rightarrow (CH_3)_3C^+ + Br^-$. Step 2 (fast): $(CH_3)_3C^+ + OH^- \rightarrow (CH_3)_3COH$. Rate = k[RBr] (first order). The carbocation is planar (sp^2), so the nucleophile attacks equally from both faces, giving retention and inversion in equal amounts: a racemic mixture. [2 + 1]

26. (a) Kolbe reaction: phenol + NaOH → sodium phenoxide; + CO_2 (4–7 atm, 400 K), then H^+ → salicylic acid. (b) $CH_3COCH_3 + H_2$ (Ni/Pt) or $NaBH_4 \rightarrow CH_3CH(OH)CH_3$. (c) Friedel–Crafts acylation: $C_6H_6 + CH_3COCl$ (anhydrous $AlCl_3$) → $C_6H_5COCH_3 + HCl$. [1 × 3]

27. (a) $NH_3 < C_2H_5NH_2 < (C_2H_5)_3N < (C_2H_5)_2NH$. (b) $RCONH_2 + Br_2 + 4NaOH \rightarrow RNH_2 + Na_2CO_3 + 2NaBr + 2H_2O$. (c) The diazonium group is resonance-stabilised by the benzene ring. [1 × 3]

28. (a) The sequence of amino acids linked by peptide bonds. (b) Secondary and tertiary structure are lost (H-bonds break) on heating or pH change, and biological activity is lost; the primary structure stays intact (e.g. coagulation of egg white). (c) DNA has deoxyribose and thymine and is double-stranded; RNA has ribose and uracil and is single-stranded. [1 × 3]

Section D — Case-based questions (4 marks each)

29. (i) Molar conductivity as concentration approaches zero. (ii) $\Lambda° = 91.0 + 426.2 - 126.4 = 390.8$ S cm^2 mol^{-1}. (iii) The degree of dissociation increases sharply on dilution. (iv) S m^{-1}. [1 × 4]

30. (i) The six carbons are in a straight chain. (ii) Gluconic acid: glucose has a –CHO group. (iii) Five –OH groups. (iv) It does not react with $NaHSO_3$ or give Schiff's test, and its pentaacetate does not react with NH_2OH. [1 × 4]

Section E — Long answer (5 marks each)

31. (a) $E = E° - (0.059/n) \log([Mg^{2+}]/[Cu^{2+}]) = 2.71 - (0.059/2) \log(10^{-3}/10^{-4}) = 2.71 - 0.0295 = 2.68$ V. [3] (b) A galvanic cell that converts the combustion energy of a fuel directly into electricity. Anode: $2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$. Cathode: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$. [2]

32. (a) $2MnO_2 + 4KOH + O_2 \rightarrow 2K_2MnO_4 + 2H_2O$; then $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$ (or electrolytic oxidation). [2] (b) $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$. [1] (c) The steady decrease in size from La to Lu due to poor shielding by 4f electrons. Consequences: Zr and Hf have similar radii, and basicity decreases from $La(OH)_3$ to $Lu(OH)_3$. [2]

33. (a)(i) Tollens' reagent: propanal gives a silver mirror, propanone does not. (ii) Iodoform test: acetophenone ($CH_3CO–$) gives a yellow CHI_3 precipitate, benzophenone does not. [2] (b) $2CH_3CHO$ (dil. NaOH) → $CH_3CH(OH)CH_2CHO$ (aldol); on heating → $CH_3CH=CHCHO$ (but-2-enal). [2] (c) $CH_3COOH < ClCH_2COOH < FCH_2COOH$. [1]